The Impact of Polysemanticity on the Locality of Neuron-level vs. Layer-level Model Editing

Hongyi Chen, Yubo Fan, Tianhao Qian, Jie Wen

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Abstract

Neuron-level model editing (e.g., FiNE) claims higher locality (fewer side effects) than layer-level methods (e.g., ROME). **However, these claims are not well-tested in 'worst-case' scenarios involving 'polysemanticity' (one neuron encoding multiple facts).** This project will systematically compare FiNE and ROME specifically on polysemantic neurons. We hypothesize that FiNE's precision will become a disadvantage in this context, causing *more* collateral damage to other facts than ROME. Our experiments on Pythia-6.9B aim to reveal a fundamental trade-off between editing precision and robustness to internal model representations.

1 Introduction and Background

- *Locating and Editing Factual Associations in GPT* (Meng et al., 2022): Introduced ROME, using Causal Tracing to find FFN layers and apply a Rank-One update.
- *Mass-Editing Memory in a Transformer* (Meng et al., 2022b): Extended ROME to support efficient batch editing of many facts at once.
- *Towards Monosemanticity...* (Bricken et al., 2023): Used noisy activations (causal intervention) to pinpoint which layer stores a fact.
- *Precise Localization of Memories...* (Hao et al., 2025): Introduced FiNE, a neuron-level method using fine-grained Causal Tracing to edit only "key neurons".
- *Toy Models of Superposition* (Elhage et al., 2022): Provided theoretical and empirical evidence for "Polysemanticity" (Superposition) in Transformers.
- *Transformer Feed-Forward Layers Are Key-Value Memories* (Geva et al., 2021): Showed that FFN layers act as "key-value memories," which inspired ROME.
- *Pythia: A Suite for Analyzing...* (Biderman et al., 2023): Released the Pythia suite, an open-source model suite ideal for mechanistic research.
- *Finding Neurons in a Haystack...* (Gurnee et al., 2023): Found that specific concepts (e.g., "Golden Gate Bridge") can be localized to individual neurons.

2 Key Questions

1. Can we develop a scalable pipeline to systematically identify 'polysemantic neurons'—those encoding multiple unrelated facts—in a large model like Pythia-6.9B?
2. (Core Question) When editing fact A, which is supported by a 'polysemantic neuron', does FiNE (neuron-level) cause more severe 'collateral damage' to fact B, also supported by that neuron, compared to ROME (layer-level)?

3 Hypothesis

Core Hypothesis: In the face of polysemanticity (Superposition), the 'precision' of neuron-level editing (FiNE) will transform from an advantage to a disadvantage.

Expectations

- When editing a 'polysemantic neuron', FiNE is expected to cause significantly more collateral damage to other facts stored in that neuron compared to ROME.
- If confirmed, this will reveal an underexplored trade-off: an edit's 'precision' vs. its 'robustness' to internal model representations like polysemanticity.

4 Methods

4.1 1. Setup

- **Model:** Pythia-6.9B
- **Tools:** ROME (layer-level) vs. FiNE (neuron-level)

4.2 2. Experiment 1: Locate Polysemantic Neurons

- Use Causal Tracing to find "key neurons" for thousands of facts.
- Use collision analysis to identify "polysemantic neurons" (storing multiple facts).
- Create two test sets: a **Polysemantic Group** (facts share neurons) and a **Clean Group** (control).

4.3 3. Experiment 2: Test Editing Locality

- On both groups, edit Fact A using ROME and FiNE.
- **Core Evaluation:** Measure accuracy drop on Fact B (collateral damage).
- **Hypothesis:** FiNE will cause more damage than ROME in the **Polysemantic Group**.